GTS-VIT: A GROUPED TOKEN SELECTIVE TRANSFORMER WITH MULTI-SCALE DUAL ATTENTION FOR HYPERSPECTRAL IMAGE CLASSIFICATION

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ABSTRACT

Recently, Vision Transformer (ViT)-based models have shown strong performance in hyperspectral image classification (HSIC) by capturing long-range dependencies and global spatial features. However, their stacked Transformer design leads to high parameter complexity, and fixed receptive fields limit adaptability to varied contextual scales. To overcome these issues, we extend the model in [1] with two key improvements. First, we introduce a Multi-Scale Dilated Attention (MSDA) module that adaptively fuses multi-receptive-field features and recalibrates channels via attention with a residual connection. Second, we integrate a token-selective strategy within a group-wise ViT framework to emphasize the most informative spatial contexts. Experiments on three benchmark datasets show that the proposed group-wise token-selective ViT achieves competitive performance, especially with limited training samples, compared to existing HSIC methods. The source code is available at: https://github.com/nedatghd/GTSViT

***Index Terms*—** MSDA, GTS-ViT, GSC-ViTgroupwise convolution, hyperspectral image classification

1. INTRODUCTION

Hyperspectral imaging systems capture hundreds of contiguous spectral bands across the electromagnetic spectrum, providing rich spectral signatures that enable fine-grained land-cover discrimination. The resulting high-dimensional data cubes, however, pose significant challenges for classification models: the curse of dimensionality, high inter-band correlation, and the scarcity of labeled training samples[2]. Recent advances in deep learning have shifted the dominant paradigm for HSIC from hand-crafted spectral-spatial features toward hierarchical learned representations. Convolutional Neural Networks (CNNs), including 1D spectral CNNs, 2D spatial CNNs, and 3D joint spectral-spatial CNNs, established strong baselines. More recently, Vision Transformer (ViT) models that adpots MSA [3] have demonstrated superior ability to model long-range spatial dependencies. However, standard ViT architectures incur quadratic complexity with respect to sequence length and typically require large labeled datasets [4]. Zhao et al.[1] proposed GSC-ViT, a lightweight hybrid CNN–Transformer architecture for hyperspectral image classification, aimed at overcoming the limited global modeling capability of CNNs and the high parameter cost and weak locality modeling of standard Vision Transformers. The method introduces three key contributions: (1) a Groupwise Separable Convolution (GSC) module that efficiently extracts local spectral–spatial features while reducing model parameters, (2) a Groupwise Separable Self-Attention (GSSA) mechanism that decomposes conventional multi-head self-attention into local and global components to lower computational complexity, and (3) a lightweight pointwise convolutional replacement for the MLP layer combined with an enhanced skip-connection scheme to facilitate effective feature fusion. Building upon GSC-ViT, The architecture proposed in this work incorporates two targeted enhancements, namely multi-scale dilated feature extraction with learned scale fusion complemented by ECA-Net–based channel recalibration, and a selective top-k attention mechanism within the Transformer module. These modifications enhance feature representation capacity while introducing only a marginal increase in model complexity.

2. METHODS

2.1. Hyperspectral Datasets

Experiments are conducted on three benchmark datasets summarized in Table 1. Indian Pines and Salinas were acquired by the AVIRIS sensor [5]; WHU-Hi-LongKou was collected via a UAV-mounted sensor over Hubei, China [6, 7, 8].

Table 1. Benchmark dataset summary.

Dataset	Size	Bands	Classes	Train	Val	Test
Indian Pines	145×145	200	16	5%	5%	90%
Salinas	512×217	204	16	1%	1%	98%
WHU-Hi-LongKou	550×400	270	9	0.2%	0.2%	99.6%

2.2. Model Architectures

As can be seen in Fig.2, the GTS-ViT follows a hierarchical pipeline that processes a raw hyperspectral patch of shape (B, C, H, W) — where B is the batch size, C is the number of spectral bands, and $H \times W$ is the spatial patch size — and

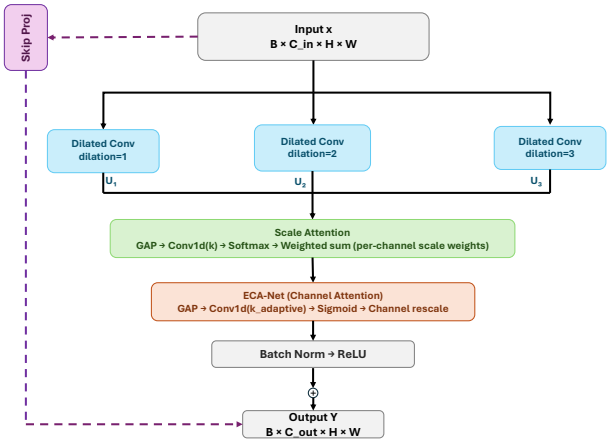

Fig. 1. MSDA — Multi-Scale Dilated Attention Module

outputs class logits. The pipeline consists of four sequential stages:

- Spectral Calibration (SC): maps the variable-band input to a fixed 256-channel feature map.
- MSDA module: extract multi-scale local spectral-spatial features at progressively reduced channel dimensions.
- GTSSA Transformer Blocks: model local and global spatial dependencies via grouped token selective self-attention.
- Classification Head: global average pooling followed by layer normalization and a linear classifier.

The network progressively compresses channel dimensionality across two stages, each consisting of one MSDA module and one GTS Transformer block.

2.2.1. *Spectral Calibration Module*

To accommodate hyperspectral datasets with varying numbers of spectral bands, the model employs a Spectral Calibration (SC) module as the initial processing stage. This module uses a 1×1 convolution followed by batch normalization and ReLU activation to project spectral dimensions to the same number of channel while preserving the spatial structure.

2.2.2. *Multi-Scale Dilated Attention (MSDA) Module*

The MSDA module employs three parallel dilated grouped convolutions, a learned scale-fusion attention mechanism, and an ECA-Net channel recalibration module [9] to enhance feature representation. Its design is motivated by the limitation of a single 3×3 convolution, which captures only a fixed local receptive field; in hyperspectral scenes, where objects vary significantly in spatial extent (e.g., narrow roads versus expansive forests), multi-scale feature extraction enables the model to simultaneously capture both fine-grained and coarse spatial patterns.

2.2.3. *Parallel Dilated Grouped Convolutions*

Three 3×3 grouped convolutions with dilation rates $d \in \{1, 2, 3\}$ are applied in parallel to the same input feature map:

$$U_i = \mathrm{GroupConv}(x; d = i, \mathrm{groups} = G), \quad i$$

Each branch uses padding equal to its dilation rate to maintain spatial resolution. G is the number of groups ($G = 8$ in practice), which reduces the parameter count relative to a standard convolution by a factor of G. Dilated convolution with rate d samples the input at intervals of d pixels, effectively expanding the receptive field of 3×3 convolution.

2.2.4. *Scale Attention*

Instead of fixed averaging or concatenation, the three feature maps are fused using a learned Scale Attention mechanism. Each scale is first compressed via global average pooling into a channel descriptor and processed by a lightweight 1D convolution to generate per-channel attention weights. These weights are normalized across scales using softmax and applied to compute a weighted sum of the feature maps, enabling adaptive receptive-field selection per channel. This design is significantly more parameter-efficient than conventional fully connected bottlenecks.

2.2.5. *ECA-Net Channel Attention*

After scale fusion, the fused feature map is further refined by Efficient Channel Attention (ECA-Net) module [9]. ECA-Net works by first summarising each channel into a single value via global average pooling, then passing these values through a small 1D convolution to capture relationships between neighbouring channels, and finally applying a sigmoid to produce a weight between 0 and 1 for each channel. These weights rescale the feature map, suppressing uninformative channels and amplifying discriminative ones.To further enhance feature propagation and stabilize training, a residual connection is introduced in the MSDA block.

2.2.6. *Transformer Architecture with Groupwise Token Selective Self-Attention (GTSSA)*

Following each MSDA block, a Groupwise Token Selective Self-Attention (GTSSA) is designed to efficiently model spatial dependencies in convolutional feature maps. Instead of applying global self-attention over all spatial tokens, the model adopts a Group Token Selective Self-Attention (GTSSA) mechanism that operates within local spatial groups and propagates contextual information through learnable group tokens. The Transformer is composed of stacked GTSSA layers arranged in a Pre-Norm residual structure, enabling effective spatial reasoning with reduced computational overhead.

2.2.7. Transformer Block

The Transformer operates on an input feature map $\mathbf{X} \in R^{B\times C\times H\times W}$ using a stack of L identical attention layers. Each layer applies channel-wise pre-normalization followed by the Group Token Selective Self-Attention (GTSSA) module, such that the output of each layer is obtained by feeding the normalized features into the attention operation. Applying normalization before attention stabilizes training and maintains consistent feature distributions, which is particularly important for deep or multi-stage Transformer architectures.

2.2.8. Group-Based Tokenization

Instead of flattening the entire feature map into a long token sequence, the Transformer divides the spatial domain into non-overlapping local groups of size $g \times g$. The input is reshaped as:

$$(B, C, H, W) \rightarrow \left(B \cdot \frac{H}{g} \cdot \frac{W}{g},\ C,\ g^2\right)$$

Each spatial group is processed independently, significantly reducing attention complexity from quadratic in HW to quadratic in g^2.To summarize each spatial group, a learnable group token is prepended to every group's token sequence. This token serves as a compact representation of the group's local context and enables later inter-group communication.

2.2.9. Selective Self-Attention within Groups

For each group, multi-head self-attention is applied using queries, keys, and values computed via 1D convolutional projections.

To further reduce redundancy, inspired by [10] a top-k selective attention strategy is employed. For each query token, only the top $k\%$ highest attention scores are retained, while the remaining scores are masked. This selective mechanism enforces sparsity and emphasizes the most relevant local interactions.In this experiment k is considered to be 0.9.

2.2.10. Inter-Group Communication via Group Tokens

When multiple spatial groups are present, the group tokens extracted from all groups are aggregated and passed through a second attention. This group-level attention allows information exchange across spatial regions without performing full global attention over all tokens.

Specifically, group tokens are reshaped into a spatial-grid sequence, new query-key representations are computed, attention weights are derived across groups, and local feature maps are aggregated accordingly. This hierarchical design enables global contextual reasoning with minimal overhead.

2.2.11. Spatial Reconstruction and Class Prediction

After attention, local group features are rearranged back to their original spatial layout, and a final convolution projects the concatenated multi-head outputs back to the original channel dimension. Using global average pooling to summarize spatial features, the model then applies a fully connected layer to output the final class prediction.

2.3. Experimental Setup and Procedure

The input feature map is initialized to $C \times 8 \times 8$, where C represents the number of spectral bands. During training, image patches are normalized to $[0, 1]$ and augmented using random horizontal or vertical flipping and rotations of $90°$, $180°$, or $270°$. The model is trained for 200 epochs using the AdamW optimizer with a learning rate of 0.001, a weight decay of 0.05, and a batch size of 128, employing cross-entropy loss. To mitigate randomness, each experiment is repeated ten times and average results are reported. Performance is evaluated using overall accuracy (OA), average accuracy (AA), and the κ coefficient. All experiments are conducted using Python 3.9 and PyTorch 2.4.1 on an NVIDIA RTX 4060 Ti GPU.

3. RESULTS AND DISCUSSION

The proposed GTS-ViT model was evaluated on three widely used hyperspectral benchmark datasets, namely Indian Pines, Salinas, and WHU-Hi-LongKou. Experimental results show that GTS-ViT achieves competitive or superior classification performance compared with several state-of-the-art CNN-based and Transformer-based hyperspectral image classification methods, as reported in [1]. Importantly, the proposed model maintains strong classification accuracy even under limited training data conditions. For example, when only 0.2% of labeled samples are used for training on the WHU-Hi-LongKou dataset, GTS-ViT achieves stable and high overall accuracy (OA) and κ coefficient values. Under this challenging setting, its performance is at least 2% higher in terms of overall accuracy than recently proposed methods such as S2CIFTNet [11], DBSSFNet [12], and GSC-ViT [1]. In addition, the low variance observed across ten repeated experiments indicates that the proposed method is robust and reliable when labeled samples are scarce, which is a common challenge in hyperspectral image classification. For the Indian Pines and Salinas datasets, GTS-ViT achieves overall accuracy (OA) and average accuracy (AA) values that are comparable to those reported for GST-ViT in [1], demonstrating consistent performance across Indian Pines and Salinas datasets. Moreover, qualitative analysis further supports these results. Visual comparisons of the classification maps shown in Figures 3–5 indicate that GTS-ViT generates predictions that are more spatially homogeneous and close to the ground-truth distributions, particularly on

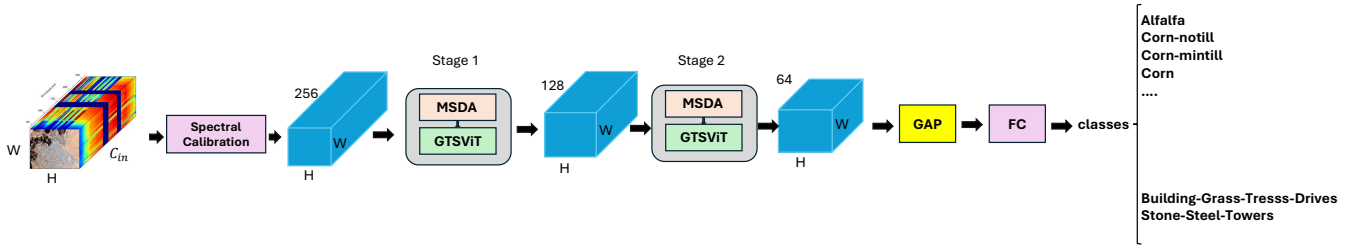

Fig. 2. Overall architecture of the proposed GTSViT framework. The network consists of spectral calibration followed by multi-stage MSDA and GTSViT blocks, and ends with global average pooling and a classification head.

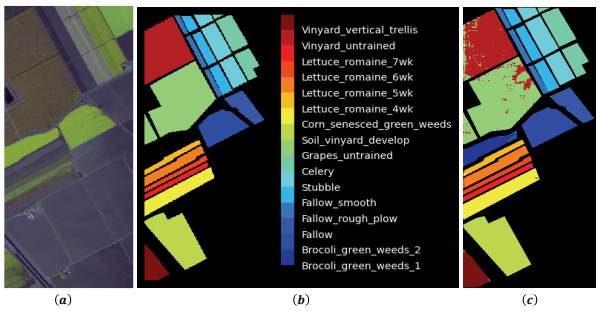

Fig. 3. False color composition of the AVIRIS Salinas scene.(b) Ground-truth map containing 16 mutually exclusive land-cover classes.(c) Classification maps obtained by the GTS-ViT model

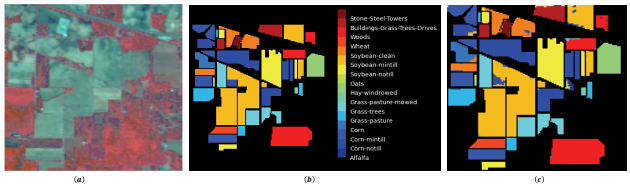

Fig. 4. False color composition of the AVIRIS Salinas scene.(b) Ground-truth map containing 16 mutually exclusive land-cover classes.(c) Classification maps obtained by the GTS-ViT model

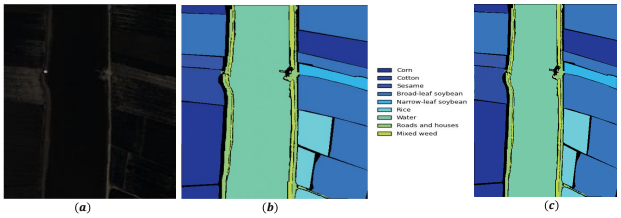

Fig. 5. (a) False color composition of the WHU-Hi-LongKou scene. (b) Ground-truth map containing nine mutually exclusive land-cover classes.(c) Classification maps obtained by the GTS-ViT model

Table 2. Classification accuracies (%) of different datasets based on GTS-ViT network.

Class	Indian Pines	Salinas	WHU-HiLongKou
1	82.93±16.61	98.54±2.74	99.95±0.05
2	94.40±2.62	99.83±0.33	99.84±0.09
3	97.62±1.75	98.57±2.08	99.68±0.37
4	92.82±6.25	98.65±1.81	99.90±0.04
5	94.99±3.27	96.62±3.61	99.06±0.60
6	98.63±1.54	100.00±0.00	99.96±0.04
7	92.80±6.40	99.89±0.21	99.98±0.01
8	99.93±0.15	92.80±1.26	98.58±0.36
9	51.11±17.53	99.94±0.17	98.76±0.34
10	93.34±2.82	98.05±1.01	–
11	97.52±0.88	95.67±4.26	–
12	93.78±3.58	98.40±2.14	–
13	99.73±0.65	99.64±0.56	–
14	98.55±1.05	99.07±1.35	–
15	93.03±5.79	90.73±3.02	–
16	93.57±3.58	97.70±3.22	–
OA (%)	96.24±0.56	96.56±0.45	**99.84±0.02**
AA (%)	92.17±1.41	97.76±0.45	**99.53±0.08**
$\kappa \times 100$	95.71±0.64	96.17±0.50	**99.79±0.03**

the WHU-Hi-LongKou dataset. Finally, several key hyperparameters, including the attention sparsity ratio k, patch size, and the number of Transformer blocks, were selected based on existing literature. Nevertheless, further experiments are required to systematically evaluate the impact of different hyperparameter settings on the model's performance.

4. CONCLUSION

This report introduced GTS-ViT, a grouped token-selective Vision Transformer for hyperspectral image classification. Built upon GSC-ViT, it incorporates two key enhancements: a Multi-Scale Dilated Attention (MSDA) module for effective multi-scale spectral–spatial feature extraction via parallel dilated convolutions with adaptive fusion and ECA-based channel recalibration, and a Groupwise Token Selective Self-Attention (GTSSA) mechanism that enables efficient sparse local-to-global spatial modeling. Experiments on Indian Pines, Salinas, and WHU-Hi-LongKou demonstrate that GTS-ViT achieves competitive or superior performance compared to recent CNN- and Transformer-based methods, particularly in scenarios with limited labeled samples.

5. REFERENCES

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